

FRAME DEMONSTRATION

SSS 1-3 | Research & Design Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for engineering design decisions.

DEMONSTRATE

Demonstrate how to frame using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-481 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

DESIGN DEMONSTRATION

SSS 1-3 | Research & Design Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for engineering design decisions.

DEMONSTRATE

Demonstrate how to design using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-482 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

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INSTRUMENT DEMONSTRATION

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Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for engineering design decisions.

DEMONSTRATE

Demonstrate how to instrument using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

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CALIBRATE DEMONSTRATION

SSS 1-3 | Research & Design Garden

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Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for engineering design decisions.

DEMONSTRATE

Demonstrate how to calibrate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

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MODEL DEMONSTRATION

SSS 1-3 | Research & Design Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for engineering design decisions.

DEMONSTRATE

Demonstrate how to model using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-485 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

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VALIDATE DEMONSTRATION

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DEMONSTRATE

Demonstrate how to validate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

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EVALUATE DEMONSTRATION

SSS 1-3 | Research & Design Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for engineering design decisions.

DEMONSTRATE

Demonstrate how to evaluate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-487 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

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OPTIMIZE DEMONSTRATION

SSS 1-3 | Research & Design Garden

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Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for engineering design decisions.

DEMONSTRATE

Demonstrate how to optimize using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

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STRESS-TEST DEMONSTRATION

SSS 1-3 | Research & Design Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for engineering design decisions.

DEMONSTRATE

Demonstrate how to stress-test using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

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RECOMMEND DEMONSTRATION

SSS 1-3 | Research & Design Garden

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Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for engineering design decisions.

DEMONSTRATE

Demonstrate how to recommend using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-490 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

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